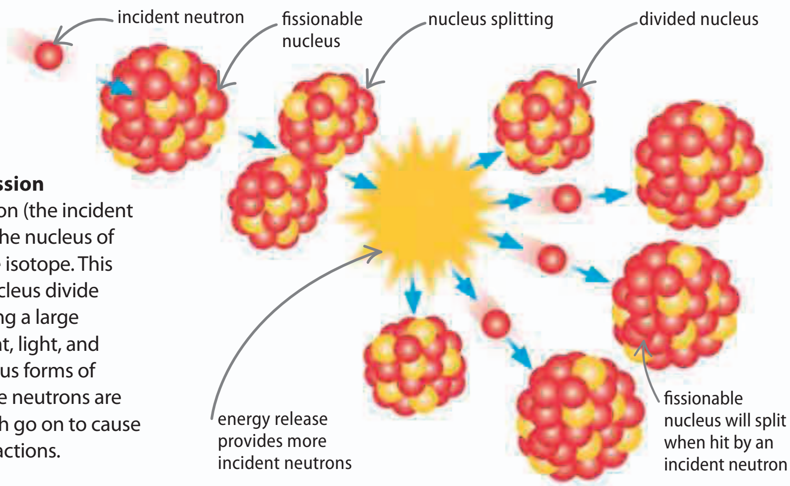
Energy from atoms

Nuclear power stations use radioactive materials, such as uranium or thorium, as a source of heat. Radioactive elements produce heat as they decay, but a great deal more heat is produced by a process called nuclear fission. The fuel is refined to contain large amounts of a particular radioactive isotope (see pages 126–127) that can be split into two smaller atoms. Uncontrolled fission causes the explosion of a nuclear bomb, but the process is slowed down in nuclear reactors.

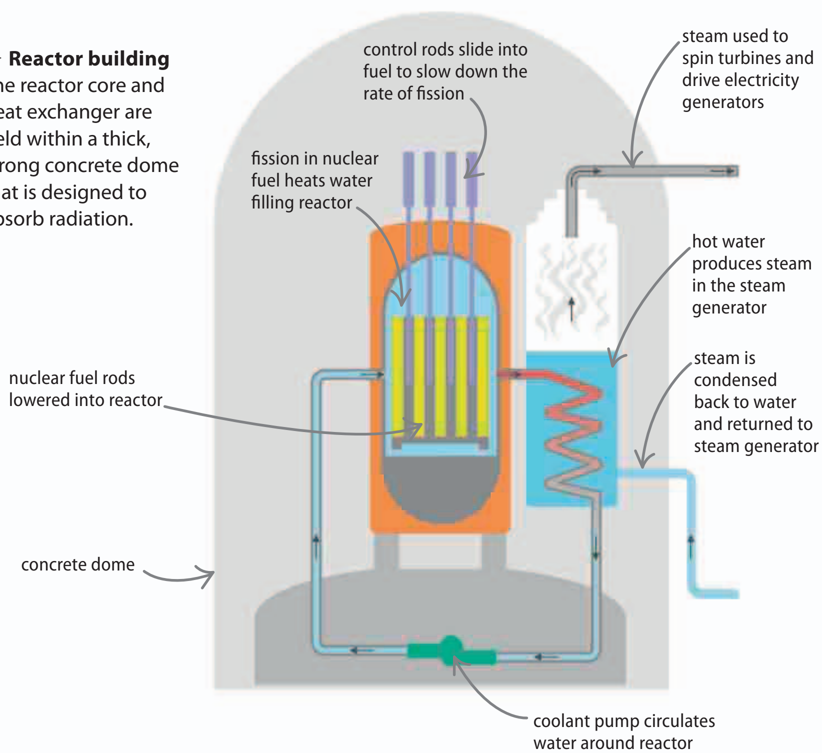
▷ **Nuclear fission**
A single neutron (the incident neutron) hits the nucleus of the fissionable isotope. This makes the nucleus divide in two, releasing a large amount of heat, light, and more dangerous forms of radiation. More neutrons are released, which go on to cause new fission reactions.



Nuclear reactor

The fission reaction takes place inside a reactor filled with water or gas. The reactor has a core containing fuel rods made of radioactive material. The reaction heats the water, which is pumped through a heat exchanger, where the superheated water makes steam that drives the turbines. There are also control rods, made largely of boron, which soak up some of the free neutrons, limiting the number of fissions that occur and so controlling the process.

▷ **Reactor building**
The reactor core and heat exchanger are held within a thick, strong concrete dome that is designed to absorb radiation.



REAL WORLD

Cherenkov radiation

The water surrounding a nuclear reactor has an eerie blue color, which is caused by Cherenkov radiation, named after the Russian scientist Pavel Cherenkov (1904–1990). This happens because charged particles move through the water at an extremely high velocity.